

Target Conference

IMC 2026 short paper

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Core Idea

We want to study when a VMNO truly competes with its host MNO, and what actually explains the performance gap from the RAN side.

Our work is a measurement study of when VMNO–host competition is actually visible at the UE side, and why CQI-based priority is only a partial explanation.

Our Argument

The VMNO-HostMNO performance hierarchy is not fixed. CQI explains some cases when using same cell, but the cell selection, RAT mode, and operator specific radio configuration can also affect the performance.

Story Line

Host MNO perform better than VMNO because Host has higher scheduling priority: partially true.

Figure 1 shows the overall performance

Senario 1: Same PCI : CQI correlates higher priority and better throughput.

Figure 2 shows performance under same PCI

Senario 2: Different PCI: No longer competing in the same scheduler context.

Figure 3 shows performance under different PCI

Senario 3: HostMNO tends to choose a worse cell.

Figure 4 shows connected PCI distribution and average throughput per PCI

Senario 4: 5G SA vs 5G NSA: VMNO with 5G NSA defeat HostMNO with 5G SA, which imply more advanced access mode does not always lead to better user-visible performance

Figure 5 shows performance SA vs NSA

Senario 5: Different MCS behavior: Under same channel conditions, the VMNO sometimes gets a much higher MCS index than the host. (Probably different MCS tables or link adaptation policies?)

Figure 6 shows VMNO got higher MCS index and correspondingly higher throughput

Takeaway

- CQI's action scope is every cell
- Cell selection can be matter
- RAT selection may affect performance
- Link adaptation configuration can be matter

Discussion

Why we use iperf throughput for statistic? - It stands for the user-visible performance.

Why we use MAC throughput for analysis?? - It reveals link behavior.

Why we think the differences are primarily RAN-Driven? - We use same iperf server. All tests have similar RTT. Performances fluctuations align with RAN-specific events.

Related Work

A First Look at Performance in Mobile Virtual Network Operators (IMC 2014) Application level performance comparison

An In-depth Study of Commercial MVNO: Measurement and Optimization (MobiSys 2019) MVNO ecosystem / architecture / optimization

Understanding Operational 5G: A First Measurement Study on Its Coverage, Performance and Energy Consumption (SIGCOMM 2020)

Performance of Cellular Networks on the Wheels (IMC 2023)

Cross-layer measurement framework.

A Peek into 5G NSA vs. SA Control Plane Performance (HotMobile 2025) 5G SA does not always outperform NSA.

Our novelty

First study of VMNO-HostMNO competition through UE-RAN link layer evidence.